

# Direct marked cancellation and range-sensitive endpoint errors

Complete auxiliary lemmas; independently reviewed

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Suppose nondecreasing comparison columns  $p_1, \dots, p_m$  and positive probability row weights  $w$  satisfy  $\mathbb{E}_w \prod_{j \in S} p_j = 1/(|S| + 1)$  for all subsets  $S$ . Write  $u_j = 1 - p_j$ . Fix a marked column  $i$ , put  $M = m - 1$ , choose  $\ell = \lfloor M/4 \rfloor$ , and put  $N = M - \ell$ . For a uniformly chosen  $\ell$ -subset  $B$  of the other  $M$  columns, let  $R$  be its complement among those  $M$  columns and set

$$A_B = \prod_{j \in B} p_j, \quad z_j = \ell u_j, \quad v_B = N^{-1} \sum_{j \in R} z_j, \quad x = M^{-1} \sum_{j \neq i} z_j, \quad h_i = z_i - x.$$

All these quantities except  $B, R$  are functions of the row. For a univariate polynomial  $P$ ,  $\text{Pol}_N(P)$  denotes its symmetric multiaffine polarization, with  $t^k$  replaced by  $e_k(z_1, \dots, z_N)/\binom{N}{k}$ .

**Lemma 1** (Exact cancellation before scalar comparison). *For every polynomial  $P$  of degree at most  $N - 1$ ,*

$$\mathbb{E}_w \mathbb{E}_B A_B [z_i \text{Pol}_N(P)(z_R) - \text{Pol}_N(tP(t))(z_R)] = 0. \quad (1)$$

For a fixed row and subset set

$$a_k = \frac{e_k((z_j - v_B)_{j \in R})}{\binom{N}{k}}.$$

Thus  $a_0 = 1, a_1 = 0$ . The same identity can be written

$$\mathbb{E}_w \mathbb{E}_B A_B (z_i - v_B) P(v_B) = \mathbb{E}_w \mathbb{E}_B A_B \sum_{k=2}^{\deg P+1} \left[ \frac{(v_B - z_i) P^{(k)}(v_B)}{k!} + \frac{P^{(k-1)}(v_B)}{(k-1)!} \right] a_k, \quad (2)$$

where derivatives above the degree are zero. The leading  $k = 2$  term uses  $a_2 = -\text{Var}_R(z)/(N-1)$ .

*Proof.* For a monomial  $P = t^r$ , every term on the left of (1) has  $\ell$  distinct success factors and  $r + 1$  distinct failure factors; the latter include the marked index on the first side and are all in  $R$  on the second. The mixed-moment identities, by expanding the failure factors, give in both cases

$$\ell^{r+1} \int_0^1 t^\ell (1-t)^{r+1} dt.$$

The index sets are disjoint and have total size at most  $m$ , since  $r + 1 \leq N$ . Linearity proves (1). The centered polarization formula is

$$\text{Pol}_N(P)(z_R) = \sum_{k=0}^{\deg P} \frac{P^{(k)}(v_B)}{k!} a_k.$$

Also  $(tP)^{(k)}(v_B) = v_B P^{(k)}(v_B) + k P^{(k-1)}(v_B)$ . Substitute these formulas into (1), use  $a_1 = 0$ , and rearrange. Finally  $e_2(y) = -\sum_j y_j^2/2$  when  $\sum_j y_j = 0$ .  $\square$

The point of (2) is the cancellation of the common scalar approximation error. There is no error term independent of the range of the columns.

**Lemma 2** (A centered coefficient bound). *Let  $H = \max_j z_j - \min_j z_j$ , including the marked column if desired. For every  $2 \leq k \leq N$ ,*

$$|a_k| \leq \left( CH \sqrt{k/N} \right)^k.$$

*Proof.* Put  $y_j = z_j - v_B$  and  $\sigma^2 = \sum_{j \in R} y_j^2 \leq NH^2$ . For every complex  $t$ , the elementary inequality  $|1 + ty| \leq \exp(\operatorname{Re}(ty) + |ty|^2/2)$  gives

$$\left| \prod_j (1 + ty_j) \right| \leq \exp(|t|^2 \sigma^2 / 2),$$

because  $\sum_j y_j = 0$ . Cauchy's coefficient estimate at radius  $\sqrt{k}/\sigma$  yields  $|e_k(y)| \leq (e\sigma^2/k)^{k/2}$ . Divide by  $\binom{N}{k} \geq (N/k)^k$ . The case  $\sigma = 0$  is immediate.  $\square$

**Lemma 3** (Replacing the weighted subset mean). *Fix a row with  $\min_{j \neq i} p_j \geq 1/2$ , and choose  $B$  with probability proportional to  $A_B$ . Suppose  $H \leq 1$ . Then, with constants absolute for  $M$  sufficiently large,*

$$0 \leq \mathbb{E}_B(v_B - x) \leq CH^2/M, \quad \mathbb{E}_B(v_B - x)^2 \leq CH^2/M.$$

*Consequently, if  $P$  is twice continuously differentiable on the interval  $I = [\min_j z_j, \max_j z_j]$ , then*

$$|\mathbb{E}_B[(z_i - v_B)P(v_B)] - h_i P(x)| \leq \frac{CH^2}{M} (\|P\|_{\infty, I} + \|P'\|_{\infty, I} + H\|P''\|_{\infty, I}). \quad (3)$$

*Proof.* The exact success-weighted subset identity in `../rational_designs/asymptotic/weighted_success_subset_moments.tex` gives

$$\mathbb{E}_B(v_B - x) = \frac{\ell}{MN} \sum_{a < b} (p_a - p_b)^2 \frac{e^{\ell-1} (p_{-[a,b]})}{e_\ell(p)}.$$

The ratio is at most two, and  $\sum_{a < b} (p_a - p_b)^2 = M \sum_a (p_a - \bar{p})^2 \leq M^2 H^2 / \ell^2$ . Since  $\ell, N$  are comparable to  $M$ , this proves the bias bound. For the variance, subtract  $\min_j z_j$  from every coefficient in the selected sum; its variance is unchanged because  $|B| = \ell$  is fixed. All remaining coefficients are between 0 and  $H$ . Pair inclusion covariances are nonpositive, as proved in the cited note, so

$$\operatorname{Var}(v_B) \leq N^{-2} \sum_{j \neq i} (z_j - \min_a z_a)^2 \leq CH^2/M.$$

Adding the squared bias proves the second-moment bound when  $H \leq 1$ . For  $g(v) = (z_i - v)P(v)$ , on  $I$  one has

$$|g'(v)| \leq H|P'(v)| + |P(v)|, \quad |g''(v)| \leq H|P''(v)| + 2|P'(v)|.$$

Taylor's theorem at  $x$ , the bias, and the second moment give (3).  $\square$

**Lemma 4** (Subgaussian subset shifts under every tilt). *In the setting of Lemma 3, put  $Y = v_B - x$ . There is an absolute constant  $C$  such that*

$$\log \mathbb{E}_B e^{t(Y - \mathbb{E}_B Y)} \leq CH^2 t^2 / M \quad (t \in \mathbb{R}).$$

*For every fixed  $C_0 > 0$ , if  $\tau \geq C_1 H^2 / M$  with a sufficiently large  $C_1 = C_1(C_0)$ , then*

$$\mathbb{E}_B e^{C_0 Y^2 / \tau} \leq C_{C_0}, \quad \mathbb{E}_B [Y^2 e^{C_0 Y^2 / \tau}] \leq C_{C_0} H^2 / M.$$

*Proof.* The exponential tilt of the weighted subset law by  $\exp(t \sum_{j \in B} z_j)$  is again a positive product-weight law on fixed-size subsets, with weights  $p_j e^{t z_j}$ . Its pair inclusion covariances are nonpositive by the same Newton-inequality proof; that argument requires positivity, not an upper bound of one on the weights. Subtracting  $\min_j z_j$  from all coefficients leaves the variance unchanged because the subset size is fixed. Thus, under every real tilt,

$$\text{Var}_t \left( \sum_{j \in B} z_j \right) \leq M H^2.$$

The second derivative of its log moment-generating function is exactly this tilted variance. Integrate the bound twice from zero and divide by  $N^2$ , using  $N \asymp M$ , to obtain the displayed subgaussian bound for  $v_B$  centered at its own mean. Lemma 3 gives  $|\mathbb{E}_B Y| \leq C H^2/M$ . Since  $H \leq 1$ , its squared mean is at most  $C H^2/M$ . The subgaussian tail estimate and integration of the tail, or the elementary Gaussian integral identity for  $e^{cY^2}$ , give the two final bounds when  $C_1$  is large enough. When  $H = 0$ ,  $Y = 0$  identically.  $\square$

**Lemma 5** (A localization-preserving subset comparison). *Retain those assumptions, take  $0 < \tau \leq 1$  with  $\tau \geq C_1 H^2/M$ , and fix a nonnegative envelope  $A(x)$ . Suppose  $P$  has, for  $r = 0, 1, 2$  and every  $y \in I$ ,*

$$|P^{(r)}(y)| \leq C_0 A(x) \tau^{-r/2} \exp(C_0(y-x)^2/\tau).$$

*Then*

$$\begin{aligned} |\mathbb{E}_B P(v_B) - P(x)| &\leq \frac{C H^2}{M \tau} A(x), \\ |\mathbb{E}_B [(z_i - v_B) P(v_B)] - h_i P(x)| &\leq \frac{C H^2}{M} (1 + \tau^{-1/2} + H \tau^{-1}) A(x). \end{aligned}$$

*The constant depends only on  $C_0$ .*

*Proof.* Use Taylor's theorem at  $x$ . The first derivative term is bounded by the bias  $C H^2/M$  times the appropriate derivative at  $x$ . For the integral second-order remainder, the displayed envelope at  $x + tY$  is bounded by its right side with  $Y^2$  in the exponential. Lemma 4 bounds its expectation, including the factor  $Y^2$ , by  $C H^2/M$ . For the second assertion apply this argument to  $g(y) = (z_i - y)P(y)$  and use

$$|z_i - y| \leq H, \quad g' = (z_i - y)P' - P, \quad g'' = (z_i - y)P'' - 2P'.$$

The stated bounds follow. This argument keeps the envelope at the original mean  $x$ , instead of replacing it by a supremum on a potentially much longer row-range interval.  $\square$

**Lemma 6** (Localized direct marked differential sum). *Fix  $0 < a_0 < L < \infty$ , and suppose all row values  $z_j$  and a kernel center  $a$  lie in  $[a_0, L]$ . Let  $K_\tau$  be the Laguerre Poisson kernel in `../localized_laguerre_kernel_derivatives.tex`. For sufficiently small  $0 < \tau < \tau_0$  (depending only on the fixed interval), if  $H \leq c$  and  $H/\sqrt{N\tau} \leq c$ , then for every  $d \leq N$*

$$\left| \sum_{k=2}^d \left[ \frac{(v_B - z_i) \partial_v^k K_\tau(v_B, a)}{k!} + \frac{\partial_v^{k-1} K_\tau(v_B, a)}{(k-1)!} \right] a_k \right| \leq C K_{4\tau}(v_B, a) \left( \frac{H^2}{N\sqrt{\tau}} + \frac{H^3}{N\tau} \right).$$

*The constants depend only on the fixed interval. Under the weighted subset law, provided  $\tau \geq C H^2/M$ , the right side averages to at most the same bound with  $K_{C'\tau}(x, a)$  and  $N$  replaced by  $M$ .*

*Proof.* The derivative-order- $k$  part is the unmarked centered polarization estimate from the cited note, multiplied by  $|v_B - z_i| \leq H$ . It is at most  $CK_{4\tau}H^3/(N\tau)$  for  $\tau \leq 1$ . For the derivative-order- $(k-1)$  part, combine that note's bound

$$\frac{|\partial_v^{k-1}K_\tau|}{(k-1)!} \leq CK_{4\tau} \left[ (C/\sqrt{(k-1)\tau})^{k-1} + C^{k-1} \right]$$

with Lemma 2. The first series is bounded by

$$CK_{4\tau} \frac{H}{\sqrt{N}} \sum_{k=2}^d \sqrt{k} (CH/\sqrt{N\tau})^{k-1} \leq CK_{4\tau} \frac{H^2}{N\sqrt{\tau}}.$$

For the second, use  $(k/N)^{k/2} \leq k/N$  and sum a geometric series in  $CH$ , obtaining  $CK_{4\tau}H^2/N$ . This proves the first assertion. The complex-shift bound in the cited kernel note, applied to the real shift  $v_B - x$ , gives

$$K_{4\tau}(v_B, a) \leq CK_{16\tau}(x, a) \exp(C(v_B - x)^2/\tau)$$

when the row range is sufficiently small. Lemma 4 then proves the weighted averaging assertion.  $\square$

As with the corresponding unmarked kernel estimate, this lemma concerns a finite differential sum. The exact polynomial identity (2) can only use it after the spectral truncation of the Poisson kernel has been quantitatively justified.

**What this does and does not prove.** Equations (2) and (3) control the marked functional  $h_i P$  by errors quadratic in the row range  $H$ . When all profiles agree, both errors vanish exactly. Combining this with a positive localized polynomial test may improve the known endpoint-profile rate. No such sharp rate is claimed here: one still has to control localization, negative contributions outside a one-sided deviation interval, and tails of the row measure.